



# Prevalence of Instagram addiction among Iranians: A countrywide online survey

Sayed Ali Sharifi Fard<sup>a,\*</sup>, Mark D. Griffiths<sup>b</sup>, Majid Jafari Abesh Ahmad Loo<sup>c</sup>, Afshin Afzali<sup>d</sup>,  
Golnaz Ali Babaei<sup>e</sup>, Mojtaba Aflaki<sup>f</sup>, Arefeh Bagher Pour<sup>g</sup>

<sup>a</sup> Faculty of Educational Sciences and Psychology, University of Mohaghegh Ardabili, Ardabil, Iran

<sup>b</sup> International Gaming Research Unit, Psychology Department, Nottingham Trent University, Nottingham, UK

<sup>c</sup> General Department of Education of Alborz Province, Alborz, Iran

<sup>d</sup> Department of Psychology, Faculty of Economic and Social Sciences, Bu Ali Sina University, Hamadan, Iran

<sup>e</sup> Faculty of Educational Sciences and Psychology, University of Mohaghegh Ardabili, Ardabil, Iran

<sup>f</sup> Faculty of Communication Sciences, Allameh Tabataba'i University, Tehran, Iran

<sup>g</sup> Islamic Azad University, Tehran Science and Research Unit, Faculty of Literature, Humanities and Social Sciences, Tehran, Iran

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## ABSTRACT

Instagram is the most popular social network platform among Iranians, especially young people. This popularity is due to several factors, including the features and attractiveness of this platform. However, recent research has indicated that for some individuals, excessive use of Instagram can be problematic and has been termed 'Instagram addiction' by some scholars. However, most previous studies (including those in Iran) have been small-scale. Therefore, a cross-sectional online survey examined the prevalence of Instagram addiction (using the Persian version of the Instagram Addiction Scale [IAS]) among Iranians (aged 18 to 50 years old) selected by multi-stage cluster sampling (by lottery) in five provincial capitals. The final sample comprised 1854 participants. The results indicated that 23.6 % were classed as having 'severe' addiction. The results also showed that IAS scores based on gender (being female) ( $p < 0.01$ ), age (being younger) ( $p < 0.01$ ), marital status (being single) ( $p < 0.01$ ), and occupational status (being a student) ( $p < 0.01$ ) were significantly different. However, there was no significant difference in IAS scores based on education and economic status ( $p > 0.05$ ). Regression analysis showed that age (being younger) and marital status (being single) were significant predictors of Instagram addiction. Given the high levels of problematic Instagram use among Iranian adults, education and prevention programs need to be implemented at a national level.

## 1. Introduction

Addiction to social media platforms is not classified or formally recognized in any medical or psychiatric texts. However, two disorders (gambling disorder and internet gaming disorder) are defined as behavioral addictions in the latest fifth edition of the *Diagnostic and Statistical Manual of Mental Disorders* (DSM-5; [1]) and the eleventh revision of the *International Classification of Diseases* (ICD-11; [2]). Both of these disorders can occur online. However, problematic social media use (e.g., addictions to Facebook or Instagram) may one day be included in such diagnostic texts.

### 1.1. Online addictions

Recent meta-analytic and systematic review studies regarding online addictions have provided important insights. A meta-analysis on internet addiction among young adults (30 studies;  $N = 21,378$ ) in Europe, Asia, America and Oceania showed that internet addiction had increased over time and that internet addiction was more prevalent among females and younger adults [3]. Another meta-analysis showed that before the COVID-19 pandemic, the prevalence rate of internet addiction was 6 % to 9.7 %, but with the start of the pandemic, spending time on the internet and the prevalence rate of internet addiction appeared to have increased [4]. In a large meta-analysis (80 studies;  $N =$

\* Corresponding author.

E-mail addresses: [sharififard68@gmail.com](mailto:sharififard68@gmail.com) (S.A. Sharifi Fard), [mark.griffiths@ntu.ac.uk](mailto:mark.griffiths@ntu.ac.uk) (M.D. Griffiths), [sahandjafariss@gmail.com](mailto:sahandjafariss@gmail.com) (M.J. Abesh Ahmad Loo), [afzali.afshin@yahoo.com](mailto:afzali.afshin@yahoo.com) (A. Afzali), [g.babaei1991@gmail.com](mailto:g.babaei1991@gmail.com) (G. Ali Babaei), [afzali.afshin@yahoo.com](mailto:afzali.afshin@yahoo.com) (M. Aflaki), [g.babaei1991@gmail.com](mailto:g.babaei1991@gmail.com) (A. Bagher Pour).

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89,281), Cheng and Li [5] reported that the global prevalence of internet addiction was 6 %. The highest prevalence was in the Middle East (10.9 %) and the lowest in Northern and Western Europe (2.6 %). In another meta-analysis of internet addiction among medical students, the pooled prevalence (10 studies;  $N = 3651$ ) was 30.1 % [6]. A meta-analysis by Modara et al. [7] (30 studies;  $N = 130,531$ ), reported a 20 % prevalence rate for internet addiction among Iranian users. This study also reported that the prevalence of internet addiction had increased in Iran from 2006 to 2015. A more recent meta-analysis by Al-Khani et al. [8] in Persian Gulf countries (9 studies; total number of participants not specified) reported that the pooled prevalence of internet addiction was 33 %. This rate was higher among females than males (48 % vs. 24 %). A meta-analytic study by Endomba et al. [9] of adolescents and emerging adults in Africa (22 studies;  $N = 13,365$ ; age range 14–26 years) reported a prevalence rate of 40.3 %.

A large meta-analysis by Cheng et al. [10] on social media addiction comprising 25 studies with 32 nationalities ( $N = 34,798$ ) reported that moderate and severe levels of social media addiction were 25 % and 13 %, respectively, and the prevalence for collectivist and individualistic countries was 31 % and 14 %, respectively. Moreover, a meta-analysis of 495 studies ( $N = 2,123,762$ ) in 64 countries by Meng et al. [11] reported prevalence rates of 17.4 % for social media addiction, and 14.22 % for internet addiction. A meta-analysis of various behavioral addictions during the COVID-19 pandemic by Alimoradi et al. [12], which examined 94 studies in 40 countries ( $N = 237,657$  young adult participants), reported that the prevalence rate for internet addiction was 10.6 %, and for social media addiction was 15.1 %. Another meta-analytic review by Salari et al. [13] that reviewed 51 studies examining social media addiction among university students ( $N = 35,520$ ), reported that the global prevalence rate was 18.4 %, and was highest in Asia (22.8 %).

Using a biopsychosocial framework, Griffiths [14] asserted that all addictions comprise six core components (i.e., salience, tolerance, withdrawal, relapse, mood modification, and conflict) including Facebook addiction and Instagram addiction [15],[16],[17]. Also, a review by Andreassen and Pallesen [18] concluded that individuals who are dependent on social media sites need to spend a lot of time on these platforms. Also, continuing to use these social media sites becomes a way to reduce negative feelings such as guilt, anxiety, restlessness, and depression. Moreover, if individuals who are dependent on social media use cannot access such sites, they become stressed, restless, and irritable, and are unable to reduce their duration of their social media use [18].

### 1.2. Instagram use

Image-based social media platforms (e.g., Instagram) have become very popular in recent years [19],[20]. Currently, Instagram is the third most used social media platform in the world behind Facebook and YouTube [21]. Instagram is an image-based social network for sharing personal photos and videos. From its inception, this platform allowed users to take photographs (typically 'selfies'), apply filters to them, chat with others who can 'like' and comment on them, and share virtual content (e.g., videos and photos) [22].

Based on statistics, the number of monthly active users of Instagram was 1.3 billion in 2023, and 1.4 billion in 2024, which is much greater compared to the 805 million users in 2018, and it is predicted that it will continue to increase until 2027 [23,24]. Also, most of the users of this platform are young people aged 18–24 years and 25–34 years [25,26]. In 2024, India and the United States of America had the highest number of Instagram users worldwide, with 413 million and 171 million active users on Instagram, respectively [21]. The percentage of Instagram users is 51.8 % for males and 48.2 % for females [23]. In Iran (where the present study was carried out), Instagram is the most popular social media platform (50.76 %), followed by Facebook (13.22 %) and Twitter (8.46 %) [27]. However, compared to mid-2022, the use of Instagram in 2023 decreased because the platform has been blocked by the Iranian government of Iran [27]. However, many Iranians still use Instagram by

using filter breakers (such as hotspots) [28].

### 1.3. Instagram addiction

Yesilyurt and Solpuk Turhan [29] found that increased time spent on Instagram was associated with social media addiction. Studies have also shown that problematic social media use (including Instagram) can have negative consequences for mental health, including depression and anxiety, and decreased psychological well-being [30],[31],[32]. Moreover, a meta-analysis and systematic review examining 18 studies ( $N = 9,269$ ) showed that the problematic social media use has a positive and significant relationship with depression, anxiety, and stress [33]. Also, extreme use of social media can lead to a decrease in the quality of couples' relationships and an increase in marital conflicts [34]. Moreover, factors such as not satisfying basic psychological needs [35], personality traits [36],[37],[38], and emotional intelligence [39], can play a role in the problematic and addictive use of social media platforms (including Instagram).

Studies have shown that problematic social media use (including Instagram) can have negative consequences for mental health, including depression and anxiety [31],[40],[32]. Moreover, studies have shown that problematic Instagram use is associated with decreased self-confidence and decreased feelings of happiness [36],[41], poor psychological well-being, and poorer mental health, including depression, anxiety, stress, and social phobia [30],[31],[32],[42]. Other studies have reported that problematic Instagram use is associated with loneliness, general anxiety, and social anxiety [43], reduced academic performance [44], fear of missing out (FOMO) in young adults [45], appearance comparison, objectification, body image and drive for thinness among females [46], stress and emotional fatigue [47], social media addiction, smartphone addiction, and internet use disorder, poor quality of life, poor quality of sleep [42], and poorer attachment to parents and peers among adolescents [48]. In relation to personality traits, problematic Instagram use has been positively associated with neuroticism and introversion, and negatively associated with agreeableness and conscientiousness [49],[50].

Problematic Instagram use has also been associated with the frequency of watching live streams, liking, being able to feel a higher sense of presence using Instagram, and using Instagram as an escape from reality [51]. Yurdağül et al. [43] reported that among male adolescents, problematic Instagram use was directly associated with loneliness, depression, general anxiety, and social anxiety, and that body image dissatisfaction partially mediated these associations. Moreover, among female adolescents, problematic Instagram use has been directly associated with depression and indirectly with general anxiety and social anxiety via body image dissatisfaction [43]. One study showed that mediating effect of problematic Instagram use between self-concept clarity and body dissociation among females [52].

There have not been many studies that have attempted to estimate the prevalence of Instagram addiction. However, Rogowska and Libera [53] in a study conducted during the COVID-19 pandemic among Polish students ( $N = 954$ ) reported a prevalence rate of 17.9 % for Instagram addiction using the six-item Bergen Instagram Addiction Scale (BIAS-6), and that females used Instagram more than males. Sharifi Fard et al. [54] in a study comprising 660 Iranian students reported that 21.7 % of participants had 'severe' addiction to Instagram using the Instagram Addiction Scale [50]. In a small sample of Indian students ( $N = 207$ ), D'Souza and Manonmani [55] reported that 1.9 % of Instagram users were addicted to it, and 10.1 % were prone to addiction using the Test for Instagram Addiction [56]. Also, Soraci et al. [42] reported that the prevalence of Instagram addiction was higher among Italian males than females ( $N = 398$ ). In a Greek study ( $N = 967$ ) it was reported that females were more dependent on Instagram than males [57], but other studies have not reported any significant differences between males and females in relation to problematic Instagram use [58],[59],[60],[16].

In a study comprising three major surveys in the United States and

the United Kingdom (total sample size = 221,096), Twenge and Martin [61] reported that teenage girls spent more time on smartphones, social media, texting, general computer use, and teenage boys spent more time playing videogames. A largescale study of over 23,000 adults in Norway also found similar results with females experiencing greater problematic social media use and males experiencing greater problematic gaming [62]. These results suggest that the association between moderate or high use of social media (e.g., Instagram) and poorer psychological well-being and mental health is generally greater for females than for males. Among both genders, heavy users of digital media are twice as likely to have mental health problems compared to low users [61]. Kose and Dugan [60] reported that more females than males used Instagram as a tool to improve their mood, to get emotional support, to reduce loneliness, to connect with friends, and to get information about their lives. The analysis also found that when the number of followers increased, the scores of social media addiction using the Social Media Addiction Scale [63] and its sub-dimensions, including mood modification, preoccupation, and relapse, also significantly increased [60]. Also, Rogowska and Libera [53] reported that the prevalence of Instagram addiction was significantly higher among females than in males.

#### 1.4. The present study

Overall, there are very few studies that have examined the prevalence of Instagram addiction, and among those that have, the sample size was generally small to medium (i.e., typically no more than 600 participants). Moreover, in relation to socio-demographic factors, most studies have only looked at gender comparisons but have not considered factors such as age, marital status, economic status, employment status, and educational status (e.g., [60],[61]). Therefore, due to the methodological shortcomings of the existing research, as well as the great popularity of Instagram among Iranians compared to other platforms [27], the purpose of the present study was to estimate the prevalence of Instagram addiction among Iranians and examine multiple demographic factors in a countrywide online survey.

## 2. Method

### 2.1. Participants and procedure

The present study used a cross-sectional online survey to collect the data. The target population was adult individuals living in Iran (aged 18 to 50 years), who were selected by multi-stage cluster sampling (by lottery) comprising five provincial capitals (Tehran, Mashhad, Urmia, Alborz, and Ahvaz). Then, using a targeted convenience sampling method, the online link to the survey was provided to the participants. Based on Cochran's formula and considering outliers and missing data, the aim was to collect data from 400 participants in each city (i.e., a total of 2000 participants). After removing outliers and those with missing data ( $n = 146$ ), the final sample for analysis 1854 participants.

A link was sent to potential participants. Individuals could only participate if they met the inclusion criteria (i.e., being an Instagram user, living in one of the five randomly selected cities, and being aged 18 to 50 years). These were the first three questions. If individuals answered 'no' to any of these three questions, they were unable to complete the survey. The exclusion criteria were being aged below 18 years or above 50 years, having a psychotic illness, and having an educational level below high school diploma. To collect data, first, in each city, 30 individuals were approached as the primary target participants. The 30 individuals were then asked to send the link of the survey to other people in that city (i.e., snowball sampling) who met the inclusion criteria. Data collection continued until 400 participants from each city had completed the survey. The web-based survey link was hosted by an Iranian company (Porsall). All participants had to give their consent before starting the survey. The Research Ethics Committee of Bu Ali Sina University approved the research protocol before

implementation.

### 2.2. Measures

**Socio-demographic information:** Data collected included the age groups of adults (18 to 30 and 31 to 50 years), gender (male and female), marital status (single, married and divorced), educational status (diploma, bachelor's degree and master's degree), economic status (low, medium and high), and occupational status (student, freelance, employed, part-time employed).

**Instagram Addiction Scale (IAS-15):** This IAS was developed by Kircaburun and Griffiths [50]; Persian version: Sharifi Fard et al. [54]. It comprises 15 items and two subscales (i.e., social effect and compulsion) and is a modification of the Internet Addiction Scale (IAT). The items (e.g., "When you are on Instagram, how often do you say to yourself 'Just a few more minutes'?" are rated on a six-point scale from 1 (never) to 6 (always) and the range of scores is between 15 and 90. The cut-off points which are based on those for the IAT are as follows: no addiction (37–15), 'mild' addiction (58–38), 'moderate' addiction (73–59) and 'severe' addiction (more than 73). In the present study, the Cronbach alphas were 0.76 for the social effect subscale, 0.81 for the compulsion subscale, and 0.87 for the whole scale.

### 2.3. Statistical analysis

Various indicators such as frequencies, percentages, means, and standard deviations were used to describe the data. IAS scores between the different socio-demographic variables were statistically compared using chi-square tests (for categorical data), and *t*-tests and ANOVAs (for ordinal data). To check the normality of the data and the homogeneity of the variance, skewness and kurtosis indices, and Leven's test was used. Tukey's post-hoc test was used to statistically test for differences where there were multiple comparisons.

## 3. Results

The socio-demographic information regarding the 1854 participants is presented in Table 1.

Table 2 shows the levels of Instagram addiction using the IAS among the Iranian population. According to the results, 22.4 % of individuals had no risk of addiction to Instagram, 29.9 % of individuals had mild addiction, 24.2 % of individuals had moderate addiction, and 23.6 %, and 2.0 % of individuals had severe addiction.

**Table 1**  
Demographic information of Instagram users (N=1854).

| Variable                   | Frequency | Percentage (%) |
|----------------------------|-----------|----------------|
| <i>Age (in years)</i>      |           |                |
| 18–30                      | 1184      | 63.9           |
| 30–50                      | 670       | 36.1           |
| <i>Gender</i>              |           |                |
| Male                       | 596       | 32             |
| Female                     | 1258      | 68             |
| <i>Marital status</i>      |           |                |
| Single                     | 653       | 35.2           |
| Married                    | 1165      | 62.8           |
| Divorced                   | 36        | 2              |
| <i>Education status</i>    |           |                |
| Diploma                    | 629       | 33.9           |
| Bachelor's Degree          | 1149      | 62             |
| Master's Degree            | 76        | 4.1            |
| <i>Economic status</i>     |           |                |
| Low                        | 229       | 12.3           |
| Medium                     | 1085      | 58.5           |
| High                       | 540       | 29.2           |
| <i>Occupational status</i> |           |                |
| Student                    | 475       | 25.7           |
| Freelance                  | 357       | 19.3           |
| Employed                   | 346       | 18.6           |
| Part-time employed         | 676       | 36.4           |

**Table 2**

Prevalence of Instagram addiction among participants (N=1854).

| Instagram addiction category | N    | Percentage | Mean  | Sd    | Minimum | Maximum | Range |
|------------------------------|------|------------|-------|-------|---------|---------|-------|
| No addiction                 | 415  | 22.4 %     | 18.31 | 3.37  | 15      | 25      | 10    |
| Mild addiction               | 554  | 29.9 %     | 35.59 | 6.73  | 26      | 48      | 22    |
| Moderate addiction           | 448  | 24.2 %     | 61.42 | 7.11  | 49      | 72      | 23    |
| Severe addiction             | 437  | 23.6 %     | 80.64 | 4.71  | 73      | 90      | 17    |
| Total                        | 1854 | 100 %      | 48.99 | 23.41 | 15      | 90      | 75    |

had severe addiction. Therefore, it could be argued that of all Instagram users in the present study, approximately 48 % of individuals ( $n = 885$ ) reported moderate to severe addiction. In Table 2, the mean score on the IAS was 48.99 ( $SD = 23.41$ ; range 15 to 90).

Skewness and kurtosis statistics were used to check the normality of data distribution. The skewness and kurtosis of the distribution were between +1 and -1 (skewness 0.15 and kurtosis -0.40), which showed that the data distribution was normal. Levene's test was performed to determine the homogeneity of variances, and no significant difference was observed in the variance of Instagram addiction scores in the sample. Given that the significance level was higher than 0.05 ( $F = 0.41$ ,  $p > 0.05$ ), the assumption of homogeneity of the variance was accepted. Next, Instagram addiction was compared according to the demographic variables of gender, age, education, occupation, marriage, and economic status. The results are presented in Table 3.

The results showed that IAS scores for gender ( $t = 4.07$ ,  $p < 0.01$ ), age ( $t = 17.57$ ,  $p < 0.01$ ), marital status ( $F = 1120.17$ ,  $p < 0.01$ ), and occupational status ( $F = 78.03$ ,  $p < 0.01$ ) were significantly different. However, there was no significant difference in IAS scores based on education and economic status ( $p > 0.05$ ). More specifically, IAS scores were significantly higher among females compared to males, and significantly higher among those aged 18–30 years compared to those aged 31 to 50 years. Tukey's post hoc test was then used to investigate the difference in IAS scores for marital status and occupational status. The results are presented in Table 4.

The results of multiple comparisons in Table 4 show that the mean IAS score among single individuals was significantly higher compared to married and divorced individuals ( $p < 0.01$ ). Also, the mean IAS score among married individuals were significantly higher compared to divorced individuals ( $p < 0.01$ ). With regards to occupational status, the results showed that the mean IAS score among students was significantly higher compared to individuals who were freelancers, employed, and part-time employed ( $p < 0.05$ ). The mean IAS score among freelance individuals was significantly higher compared to individuals who were part-time employed and employed ( $p < 0.01$ ). Finally, the mean IAS score among individuals who were employed was significantly higher

**Table 4**

The results of multiple comparisons (N=1854).

| Group (1)  | Group (2)          | Mean Difference (1–2) | St. Error | p     |
|------------|--------------------|-----------------------|-----------|-------|
| Single     | Married            | 36.32*                | 0.77      | 0.001 |
|            | Divorced           | 12.43*                | 2.69      | 0.001 |
| Married    | Divorced           | -23.88*               | 2.66      | 0.001 |
| Student    | Part-time employed | -8.15*                | 1.32      | 0.001 |
|            | Freelance          | 13.34*                | 1.54      | 0.001 |
|            | Employed           | 4.10*                 | 1.56      | 0.043 |
| Freelance  | Part-time employed | -21.49*               | 1.44      | 0.001 |
|            | Employed           | -9.23*                | 1.66      | 0.001 |
| Employment | Part-time employed | -12.26*               | 1.46      | 0.001 |

Mean differences were all significant at the  $p < 0.05$  level,

compared to individuals who were part-time employed ( $p < 0.01$ ). Multiple regression analysis was conducted to determine the predictive influence of demographic variables of gender, age, education, occupation, marital status, and economic status on Instagram addiction. The results are presented in Table 5.

The results showed that the age (being younger) and marital status (being single) emerged as significant predictors of Instagram addiction (age;  $\beta = -0.09$ ,  $p < 0.01$ , marital status;  $\beta = 0.62$ ,  $p < 0.01$ ). Other

**Table 5**

Predictors of Instagram addiction (N = 1854).

| Variables           | Instagram addiction |                 |          |
|---------------------|---------------------|-----------------|----------|
|                     | B                   | $\beta$         | p        |
| Age                 | -4.37               | -0.090          | <0.001** |
| Gender              | -0.46               | -0.009          | 0.631    |
| Marital status      | 28.43               | 0.621           | <0.001** |
| Education status    | -0.36               | -0.008          | 0.634    |
| Economic status     | 0.50                | 0.013           | 0.453    |
| Occupational status | 0.11                | 0.006           | 0.759    |
|                     | $R^2 = 0.45$        | $AdjR^2 = 0.44$ |          |

Note: B = unstandardized coefficients;  $\beta$  = standardized coefficients; \*\* $p < 0.01$ .

**Table 3**

Comparing Instagram addiction based on demographic components.

| Variable            | —                  | n    | Mean  | SD    | F (t)   | p     |
|---------------------|--------------------|------|-------|-------|---------|-------|
| Age (in years)      | 18–30              | 1184 | 55.64 | 22.51 | 17.57   | 0.001 |
|                     | 31–50              | 670  | 37.23 | 20.12 |         |       |
| Gender              | Male               | 596  | 50.84 | 23.26 | 4.07    | 0.001 |
|                     | Female             | 1258 | 46.36 | 23.38 |         |       |
| Marital status      | Single             | 653  | 72.04 | 11.16 | 1120.17 | 0.001 |
|                     | Married            | 1165 | 35.72 | 17.36 |         |       |
|                     | Divorced           | 36   | 59.61 | 24.01 |         |       |
| Education status    | Diploma            | 629  | 46.06 | 23.21 | 2.63    | 0.068 |
|                     | Bachelor's Degree  | 1149 | 47.96 | 23.55 |         |       |
|                     | Master's Degree    | 76   | 45.54 | 21.22 |         |       |
| Economic status     | Low                | 229  | 50.61 | 23.62 | 1.87    | 0.153 |
|                     | Medium             | 1085 | 48.22 | 23.32 |         |       |
|                     | High               | 540  | 48.79 | 23.21 |         |       |
| Occupational status | Student            | 475  | 49.35 | 24.47 | 78.03   | 0.001 |
|                     | Freelance          | 357  | 36.01 | 18.12 |         |       |
|                     | Employed           | 346  | 45.24 | 21.59 |         |       |
|                     | Part-time employed | 676  | 57.50 | 22.43 |         |       |

Type of test: t, F, significance level  $p < 0.05$ .



variables such as gender, education, occupation and economic status were not significant predictors of Instagram addiction ( $p > 0.05$ ).

#### 4. Discussion

The results of the present study indicated that using the Persian Instagram Addiction Scale (IAS), 22.4 % of Iranian participants were not addicted to Instagram, 29.9 % had 'mild' Instagram addiction, 24.2 % had 'moderate' Instagram addiction, and 23.6 % had 'severe' Instagram addiction. In comparison to a recent Iranian study by Sharifi Fard et al. [54] with students from Mashhad and also using the IAS, they reported that 23.2 % were mildly addicted to Instagram, 53 % were moderately addicted to Instagram and 21.7 % were severely addicted to Instagram. The prevalence of Instagram addiction was much higher in Iran than the study in India by D'Souza and Manonmani [55] who reported that 1.9 % of Instagram users were addicted, and 10.1 % were prone to addiction. However, this study had a very small sample size and used a different screening instrument.

In a study conducted by Rogowska and Libera [53] among Polish students, the results showed that the prevalence of Instagram addiction during the COVID-19 pandemic was 19.17 % using the six-item Bergen Instagram Addiction Scale. Moreover, a large meta-analysis on social media addiction (which will likely have included addiction to Instagram) comprising 32 nationalities ( $N = 34,798$ ) showed that moderate and severe levels of social media addiction were 25 % and 13 %, and the prevalence for collectivist and individualistic countries was 31 % and 14 % [10]. Given the collectivist culture in Iran, and that collectivist countries appear to have higher prevalence of social media addiction, the findings of the present study appear in line with this. More generally, internet addiction has been estimated from 7 % to 40 % in various studies [63],[65],[9],[66],[67],[68]. Findings of large meta-analyses also show similar results [4],[5],[9],[3],[6].

The high level of Instagram addiction among Iranian users can be explained from several perspectives. For example, Iranian culture is collectivist, and a previous meta-analysis showed that social media addiction is much higher in collectivist countries [10]. Other psychosocial and cultural factors may also be another reason for the high use of Instagram. High visual and auditory attractions, which create opportunities to exchange images and video calls, generally increase interpersonal interactions [69],[70]. Therefore, in Iran, creating opportunities for advertising and business, interpersonal interactions, making friends, and getting to know new people are among the factors that increase the use of Instagram. Statistics show that Iranian social media users are significantly more interested in Instagram than other social media platforms. For example, 50.76 % of social media users in Iran use Instagram, 13.22 % use Facebook, and 8.46 % use Twitter [27]. The higher rates of Instagram use may partly explain the high rate of Instagram addiction although the present study did not ask participants about their use of other social media.

On the other hand, considering the unemployment rate or jobs with low income in Iran, Instagram provides opportunities for internet businesses. Using Instagram can be a very good business opportunity, especially for individuals who do not have the possibility to earn money from other sources, such as creating a physical store or a physical clinic. Therefore, many Iranian users spend a lot of time on Instagram so that they can promote their own or others' goods and services by increasing their followers and earning money in this way. Also, due to the high use of Instagram by Iranians [71],[27], many business owners such as consultants, lawyers, or goods sellers need to be active and spend time on Instagram in order to attract followers through advertising.

It also appears that one of the other important factors explaining high Instagram use in Iran is following the news and information published on the platform. Research conducted by Rouzbahani and Tajik Esmaeili [72] showed that the form and type of information, the features of the Instagram platform, speed and accuracy, honesty, the presence of famous individuals, the lack of dependence of news on government

organizations and institutions, and the freshness of news on Instagram compared to the national media were among the most effective factors in following and trusting the news published through Instagram. In general, many of the things present on Instagram (e.g., entertainment, convenience, attractiveness, passing time, self-expression, social interaction, interpersonal relationships and information exchange) can lead to spending more time on this specific social media platform [73].

The results of a recent Iranian study showed that (i) not satisfying psychological needs (including autonomy, relatedness and, competence) was associated with Instagram addiction, and (ii) Instagram addiction was associated with a decrease in psychological well-being [35]. Therefore, not satisfying the psychological needs of users can be another reason for excessive or problematic use of Instagram. Social cultural factors have played a very effective role in not satisfying psychological needs [74],[75]. It seems that underestimating or ignoring individual's successes by families and society (other people and government institutions) in Iran can lead to unsatisfied sense of competence. Also, being dependent and not fully expressing oneself (including the expression of wants and feelings) in the family (due to cultural and economic issues) for individuals in Iran can lead to unsatisfied or incomplete satisfaction of their sense of autonomy. However, both of the stated conditions have improved somewhat in the past decade in Iran, and this improvement is largely related to the spread of social media platforms and the large amount of information now being disseminated.

The results of the present study showed that there was a significant difference in IAS scores based on age (being younger), gender (being female), marital status (being single), and occupational status (being a student). Moreover, the regression analysis showed that age and marital status were significant predictors of IA. The findings indicated that Instagram addiction was more prevalent among those aged 18–30 years, females, and single individuals. These results are in line with studies by Mousavi [76], and Rogowska and Libera [53]. Yesilyurt and Solpuk Turhan [29] also reported that females spend more time on Instagram than males. Also, there were no significant differences in IAS scores based on education and economic status. This is inconsistent with the findings of Mousavi [75] and D'Souza [58]. On the other hand, Soraci et al. [42] reported that the prevalence of Instagram addiction was higher among males than females. The present study's results concerning occupational status also showed that some groups had significantly higher IAS scores including (i) students compared to all other occupational groups; (ii) self-employed individuals compared to freelancers and civil servants; and (iii) civil servants compared to freelancers. No previous study has ever examined Instagram addiction and occupational status.

There are many factors which may explain these novel findings. First, individuals in early adulthood (18 to 30 years old), enter a transformation period which includes seeking excitement, developing interpersonal relationships, and increasing social interactions and friendships. Second, individuals in this age group are less likely to be married and/or have their own families which means they have more time for social media use (e.g., Instagram use). They are also more likely to use social media sites (e.g., Instagram), because the capacities and characteristics of the Instagram platform are attractive to youth (e.g., social interaction features, forming interpersonal friendships, dating) [26]. Being older aged and having job and/or a family limits the opportunity to use Instagram excessively.

The higher IAS scores among married individuals compared to those who are divorced might be because divorced individuals have more difficult and busy work lives because of being single, especially for women and parents who take care of the children. Because of these extra demands they would have much less time to spend on social media (including Instagram) [77],[78],[79],[80],[81]. Divorce can also lead to poor mental health (e.g., depression) [79],[82],[32],[83]. All the aforementioned reasons limit the opportunity to engage excessively with social media sites (e.g., Instagram).

There are several reasons why females may have had significantly

higher IAS scores than males. Compared to males, females are more interactive, have more social relationships, are more emotional, and more talkative [78],[84]. These characteristics increase the probability of using Instagram. Also, a high proportion of Iranian females are housewives [85] who may have more opportunity and time to use Instagram than males. Anecdotally, it appears that Iranian female social media users are active in the field of advertising and blogging.

Regarding the findings related to occupational groups, university students are usually in the 18 to 30 years age group and are single. Social media use is more attractive for this age group [26] and they also have the opportunity to spend more time on Instagram due to less responsibilities than older adults (aged 31 to 50 years) (for example, marriage and children). In addition, social media sites (e.g., Instagram, LinkedIn) are among the ways of exchanging scientific-professional information and interpersonal networking interactions. Also, anecdotally, some students create a personal page about their academic field and/or things they are interested in, post stories, follow other individuals' content, and leave comments for them.

Freelance workers, compared to part-time employed and employed, have less rules and a more flexible time frame to do their work. Therefore, they may have more opportunities to spend time on social media that can lead to addiction (e.g., [4],[50]). Also, due to Instagram's features for promoting and introducing personal businesses, selling and presenting goods and services online, and because it is personal work, there is motivation for more job activity. Individuals who are freelancers may spend more time on Instagram than part-time employed and employed individuals. Also, most employed individuals in Iran work during office hours from 8 am to 2 pm, so the of their time is their free time, and one of the leisure activities of Iranian individuals is spending time on Instagram. Freelancers may also be bloggers or engaged in providing services or goods on Instagram (because of the nature of their job) so may also spend a lot of time on Instagram.

#### 4.1. Limitations, future research, and conclusion

Although the present study had a relatively large sample size, it has a number of limitations. The sample only comprised adults aged 18 to 50 years so the results cannot be generalized to adolescents or older adults. Future research should examine these other cohorts. Also, participants were only recruited from five cities in Iran. Therefore, the sample is not representative of all Iranians in this age group and future research should sample from right across Iran. Due to the heavy criticism of IAT (the IAS is adapted from IAT), there is a possibility that this measure overestimates IA, therefore, the results should be considered with caution. Moreover, the study comprised self-report data and is therefore subject to various methods biases. Therefore, future studies should use other measures related to Instagram use such as behavioral tracking data.

Based on the results, approximately one-fifth of Iranian Instagram users have a severe Instagram addiction. To reduce addiction and prevent its increase, educational solutions should be considered including (i) education concerning the harms caused by excessive and addictive use of Instagram, and healthy ways of using Instagram (for example, by setting strict time limits for Instagram use) (ii) creating more recreation and sports facilities to encourage the general public to engage in physical activity and being in the outdoor environment [86], (iii) enhancing interpersonal relationships and their quality in the real world (for example, through communication skills training), and (iv) satisfying basic psychological needs through improvement of relationships between family members and increasing interpersonal interactions and conversations at home [35].

#### Ethical considerations

The project was approved by the ethics committee of Bu Ali Sina University (ID: IR.BASU.REC.1402.004). Also, all participants were aware of the goals and nature of the study, and participated in the study voluntarily and provided informed consent.

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#### CRediT authorship contribution statement

Griffiths is the supervisor, Dr. Afzali is the project administration, and I am the conceptualization. **Sayed Ali Sharifi Fard:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Conceptualization. **Mark D. Griffiths:** Writing – review & editing, Supervision, Methodology. **Majid Jafari Abesh Ahmad Loo:** Resources, Investigation. **Afshin Afzali:** Formal analysis, Validation, Project administration. **Golnaz Ali Babaei:** Validation, Writing – original draft. **Mojtaba Aflaki:** Resources, Visualization. **Arefeh Bagher Pour:** Methodology, Formal analysis.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### Data availability

Data will be made available on reasonable request to the corresponding author.

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